

Unit 6 - Boiling and Condensation

- Pool boiling - Boiling curve
- Hysteresis in the boiling curve
- Mechanism of nucleate boiling
- Forced convection boiling - Brief overview of internal forced convection boiling
- Condensation: Physical mechanisms
- Types of condensation
- Factors affecting condensation

Boiling

- From thermodynamics that when the temperature of a liquid at a specified pressure is raised to the saturation temperature T_{sat} at that pressure, boiling occurs.
- Likewise, when the temperature of a vapor is lowered to T_{sat} , condensation occurs.
- Boiling is a liquid-to-vapor phase change process just like evaporation, but there are significant differences between the two.
- Evaporation occurs at the liquid–vapor interface when the vapor pressure is less than the saturation pressure of the liquid at a given temperature.
- Water in a lake at 20°C, for example, will evaporate to air at 20°C and 60 percent relative humidity since the saturation pressure of water at 20°C is 2.3 kPa and the vapor pressure of air at 20°C and 60 percent relative humidity is 1.4 kPa.
- Examples of evaporation: drying of clothes, fruits, and vegetables; the evaporation of sweat to cool the human body; and the rejection of waste heat in wet cooling towers.
- Evaporation involves no bubble formation or bubble motion.

Boiling

- A liquid-to-vapor phase change process is called evaporation if it occurs at a liquid–vapor interface and boiling if it occurs at a solid–liquid interface.
- Boiling occurs when a liquid is brought into contact with a surface at a temperature above the saturation temperature of the liquid.
- Boiling, on the other hand, occurs at the solid–liquid interface when a liquid is brought into contact with a surface maintained at a temperature T_s sufficiently above the saturation temperature T_{sat} of the liquid.
- At 1 atm, for example, liquid water in contact with a solid surface at 110°C will boil since the saturation temperature of water at 1 atm is 100°C .
- The boiling process is characterized by the rapid formation of vapor bubbles at the solid–liquid interface that detach from the surface when they reach a certain size and attempt to rise to the free surface of the liquid.
- Boiling is a complicated phenomenon because of the large number of variables involved in the process and the complex fluid motion patterns caused by the bubble formation and growth.

Boiling

- As a form of convection heat transfer, the boiling heat flux from a solid surface to the fluid is expressed from Newton's law of cooling as
- Where ΔT_{excess} is called the excess temperature, which represents the excess of the surface above the saturation temperature of the fluid.

$$\dot{q}_{\text{boiling}} = h(T_s - T_{\text{sat}}) = h\Delta T_{\text{excess}} \quad (\text{W/m}^2)$$

Mechanism of boiling

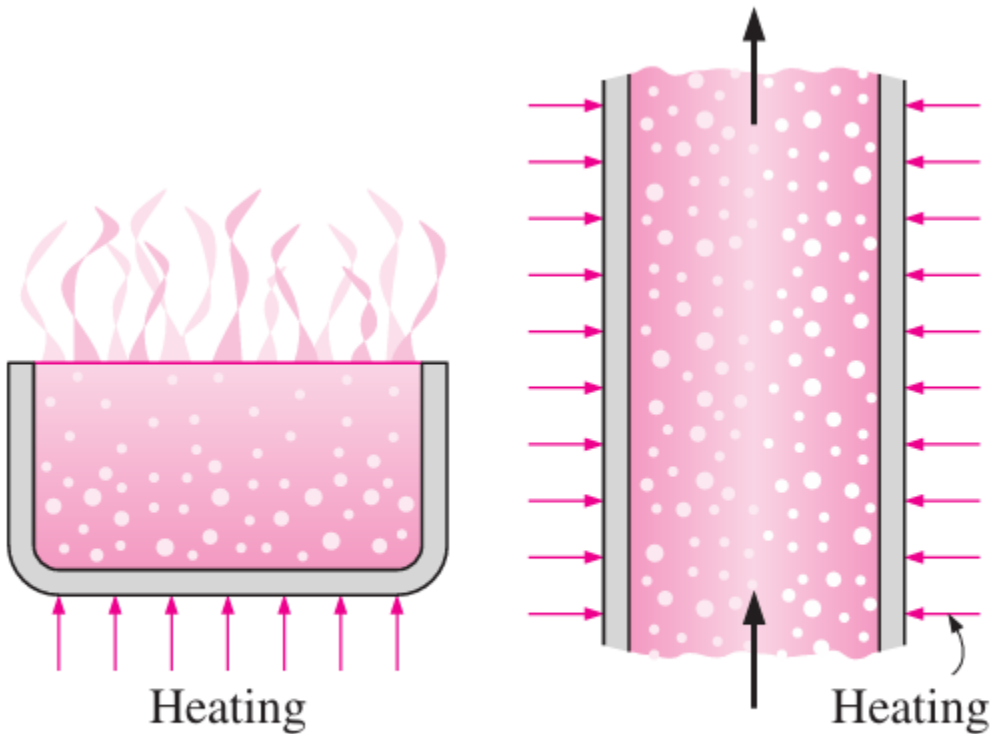
- The boiling processes in practice do not occur under equilibrium conditions, and normally the bubbles are not in thermodynamic equilibrium with the surrounding liquid.
- That is, the temperature and pressure of the vapor in a bubble are usually different than those of the liquid. The pressure difference between the liquid and the vapor is balanced by the surface tension at the interface.
- The temperature difference between the vapor in a bubble and the surrounding liquid is the driving force for heat transfer between the two phases.
- When the liquid is at a lower temperature than the bubble, heat will be transferred from the bubble into the liquid, causing some of the vapor inside the bubble to condense and the bubble to collapse eventually.
- When the liquid is at a higher temperature than the bubble, heat will be transferred from the liquid to the bubble, causing the bubble to grow and rise to the top under the influence of buoyancy.

Types of Boiling

- Boiling is classified as pool boiling or flow boiling, depending on the presence of bulk fluid motion.
- Boiling is called pool boiling in the absence of bulk fluid flow and flow boiling (or forced convection boiling) in the presence of it.
- In pool boiling, the fluid is stationary, and any motion of the fluid is due to natural convection currents and the motion of the bubbles under the influence of buoyancy.
- The boiling of water in a pan on top of a stove is an example of pool boiling.
- Pool boiling of a fluid can also be achieved by placing a heating coil in the fluid.
- In flow boiling, the fluid is forced to move in a heated pipe or over a surface by external means such as a pump.
- There fore, flow boiling is always accompanied by other convection effects.

Types of Boiling

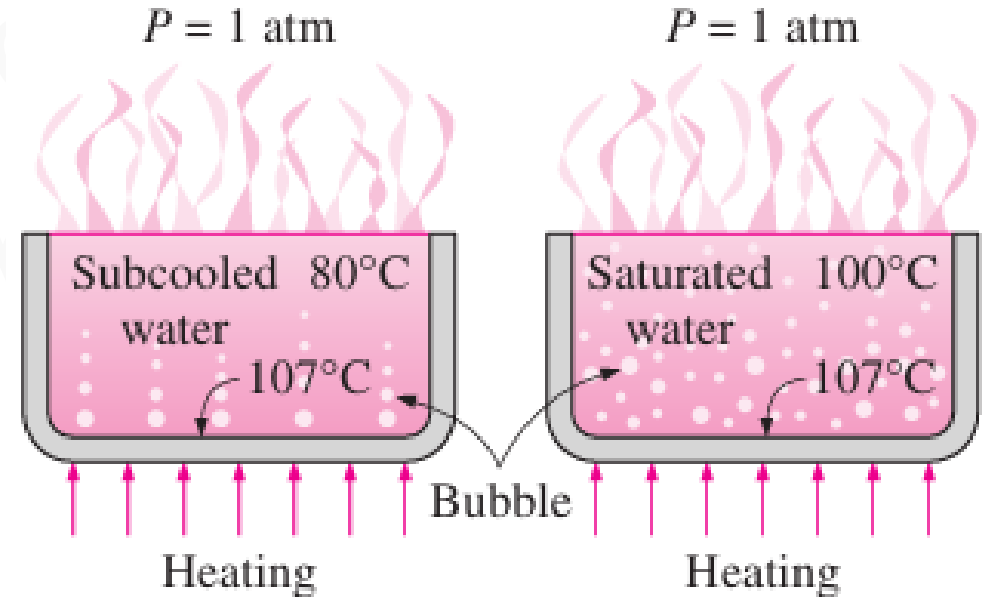
Classification of boiling on the basis of the presence of bulk fluid motion:



(a) Pool boiling

(b) Flow boiling

Classification of boiling on the basis of the presence of bulk liquid temperature:



(a) Subcooled boiling

(b) Saturated boiling

Types of Boiling

- Pool and flow boiling are further classified as subcooled boiling or saturated boiling, depending on the bulk liquid temperature.
- Boiling is said to be subcooled (or local) when the temperature of the main body of the liquid is below the saturation temperature T_{sat} (i.e., the bulk of the liquid is subcooled) and saturated (or bulk) when the temperature of the liquid is equal to T_{sat} (i.e., the bulk of the liquid is saturated).
- At the early stages of boiling, the bubbles are confined to a narrow region near the hot surface.
- This is because the liquid adjacent to the hot surface vaporizes as a result of being heated above its saturation temperature.
- But these bubbles disappear soon after they move away from the hot surface as a result of heat transfer from the bubbles to the cooler liquid surrounding them.

Types of Boiling

- This happens when the bulk of the liquid is at a lower temperature than the saturation temperature.
- The bubbles serve as “energy movers” from the hot surface into the liquid body by absorbing heat from the hot surface and releasing it into the liquid as they condense and collapse.
- Boiling in this case is confined to a region in the locality of the hot surface and is appropriately called local or subcooled boiling.
- When the entire liquid body reaches the saturation temperature, the bubbles start rising to the top.
- We can see bubbles throughout the bulk of the liquid, and boiling in this case is called the bulk or saturated boiling.

Pool boiling - Boiling curve

- In pool boiling, the fluid is not forced to flow by a mover such as a pump, and any motion of the fluid is due to natural convection currents and the motion of the bubbles under the influence of buoyancy.
- As a familiar example of pool boiling, consider the boiling of tap water in a pan on top of a stove. The water will initially be at about 15°C , far below the saturation temperature of 100°C at standard atmospheric pressure.
- At the early stages of boiling, you will not notice anything significant except some bubbles that stick to the surface of the pan. These bubbles are caused by the release of air molecules dissolved in liquid water and should not be confused with vapor bubbles.
- As the water temperature rises, you will notice chunks of liquid water rolling up and down as a result of natural convection currents, followed by the first vapor bubbles forming at the bottom surface of the pan.
- These bubbles get smaller as they detach from the surface and start rising, and eventually collapse in the cooler water above. This is subcooled boiling since the bulk of the liquid water has not reached saturation temperature yet.
- The intensity of bubble formation increases as the water temperature rises further, and you will notice waves of vapor bubbles coming from the bottom and rising to the top when the water temperature reaches the saturation temperature (100°C at standard atmospheric conditions).
- This full scale boiling is the saturated boiling.

Boiling Regimes and the Boiling Curve

- The pioneering work on boiling was done in 1934 by S. Nukiyama, who used electrically heated nichrome and platinum wires immersed in liquids in his experiments.
- Nukiyama noticed that boiling takes different forms, depending on the value of the excess temperature T_{excess} .
- Four different boiling regimes are observed: natural convection boiling, nucleate boiling, transition boiling, and film boiling.
- Boiling curve is a plot of boiling heat flux versus the excess temperature.
- Although the boiling curve given is for water, the general shape of the boiling curve remains the same for different fluids.
- The specific shape of the curve depends on the fluid–heating surface material combination and the fluid pressure, but it is practically independent of the geometry of the heating surface.

Typical boiling curve for water at 1 atm pressure

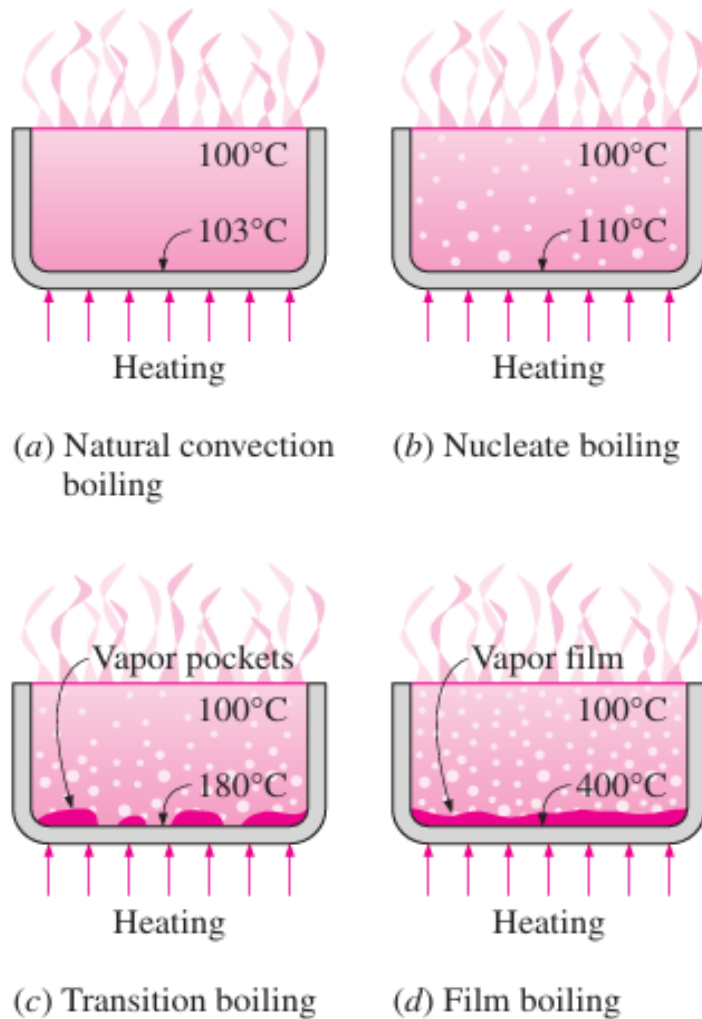
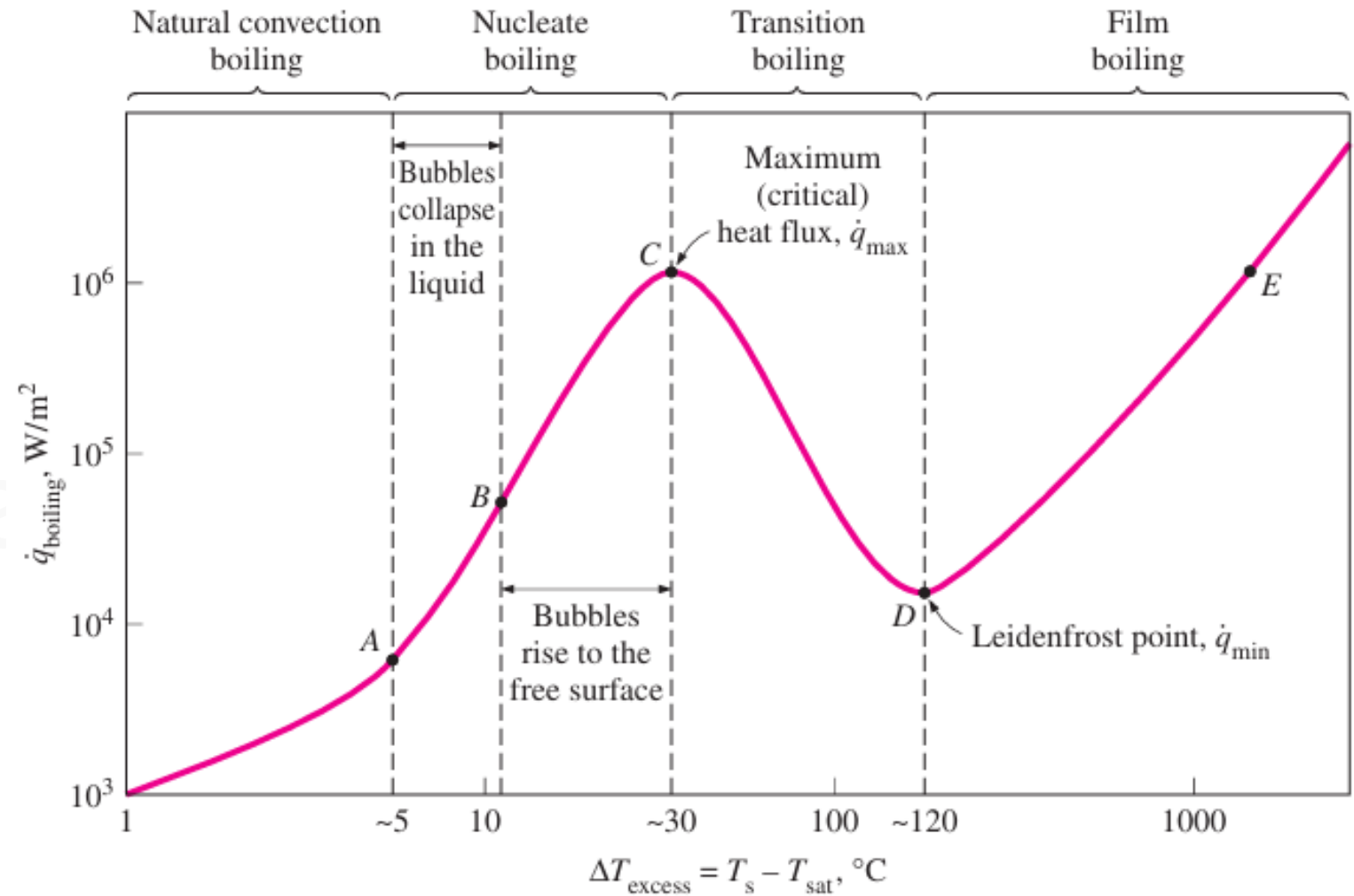


FIGURE 10-5

Different boiling regimes in pool boiling.



Boiling Regimes and the Boiling Curve

Natural Convection Boiling (to Point A on the Boiling Curve)

- We learned in thermodynamics that a pure substance at a specified pressure starts boiling when it reaches the saturation temperature at that pressure.
- But in practice we do not see any bubbles forming on the heating surface until the liquid is heated a few degrees above the saturation temperature (about 2 to 6°C for water).
- Therefore, the liquid is slightly superheated in this case (a metastable condition) and evaporates when it rises to the free surface.
- The fluid motion in this mode of boiling is governed by natural convection currents, and heat transfer from the heating surface to the fluid is by natural convection.

Boiling Regimes and the Boiling Curve

Nucleate Boiling (between Points A and C)

- The first bubbles start forming at point A of the boiling curve at various preferential sites on the heating surface.
- The bubbles form at an increasing rate at an increasing number of nucleation sites as we move along the boiling curve toward point C.
- The nucleate boiling regime can be separated into two distinct regions.
- In region A–B, isolated bubbles are formed at various preferential nucleation sites on the heated surface.
- But these bubbles are dissipated in the liquid shortly after they separate from the surface.
- The space vacated by the rising bubbles is filled by the liquid in the vicinity of the heater surface, and the process is repeated.

Boiling Regimes and the Boiling Curve

Nucleate Boiling (between Points A and C)

- The stirring and agitation caused by the entrainment of the liquid to the heater surface is primarily responsible for the increased heat transfer coefficient and heat flux in this region of nucleate boiling.
- In region B–C, the heater temperature is further increased, and bubbles form at such great rates at such a large number of nucleation sites that they form numerous continuous columns of vapor in the liquid.
- These bubbles move all the way up to the free surface, where they break up and release their vapor content.
- The large heat fluxes obtainable in this region are caused by the combined effect of liquid entrainment and evaporation.

Boiling Regimes and the Boiling Curve

Transition Boiling (between Points C and D on the Boiling Curve)

- As the heater temperature and thus the T_{excess} is increased past point C, the heat flux decreases.
- This is because a large fraction of the heater surface is covered by a vapor film, which acts as an insulation due to the low thermal conductivity of the vapor relative to that of the liquid.
- In the transition boiling regime, both nucleate and film boiling partially occur. Nucleate boiling at point C is completely replaced by film boiling at point D.
- Operation in the transition boiling regime, which is also called the unstable film boiling regime, is avoided in practice.
- For water, transition boiling occurs over the excess temperature range from about 30°C to about 120°C .

Boiling Regimes and the Boiling Curve

Film Boiling (beyond Point D)

- In this region the heater surface is completely covered by a continuous stable vapor film.
- Point D, where the heat flux reaches a minimum, is called the Leidenfrost point, in honor of J. C. Leidenfrost, who observed in 1756 that liquid droplets on a very hot surface jump around and slowly boil away.
- The presence of a vapor film between the heater surface and the liquid is responsible for the low heat transfer rates in the film boiling region.
- The heat transfer rate increases with increasing excess temperature as a result of heat transfer from the heated surface to the liquid through the vapor film by radiation, which becomes significant at high temperatures.
- A typical boiling process will not follow the boiling curve beyond point C, as Nukiyama has observed during his experiments.

Boiling Regimes and the Boiling Curve

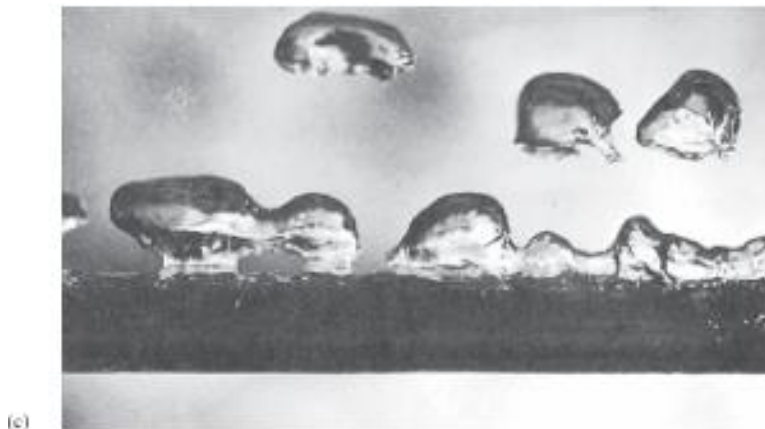
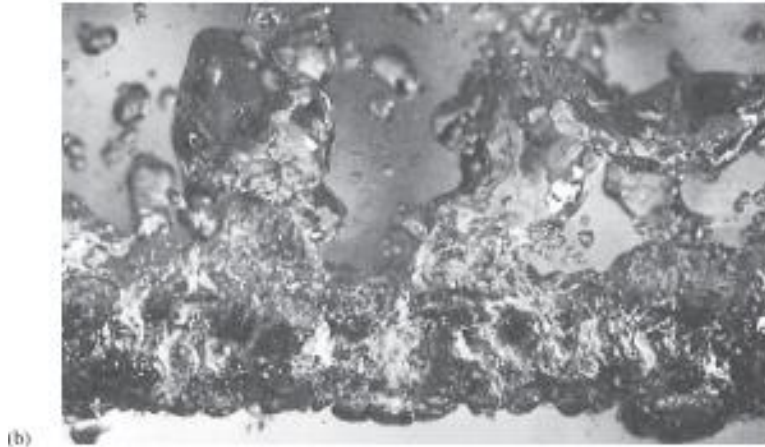
Film Boiling (beyond Point D)

- The burnout phenomenon in boiling can be explained as follows: In order to move beyond point C where $q \cdot \max$ occurs, we must increase the heater surface temperature T_s . To increase T_s , however, we must increase the heat flux.
- But the fluid cannot receive this increased energy at an excess temperature just beyond point C.
- Therefore, the heater surface ends up absorbing the increased energy, causing the heater surface temperature T_s to rise.
- But the fluid can receive even less energy at this increased excess temperature, causing the heater surface temperature T_s to rise even further.
- This continues until the surface temperature reaches a point at which it no longer rises and the heat supplied can be transferred to the fluid steadily.

Boiling Regimes and the Boiling Curve

Film Boiling (beyond Point D)

- This is point E on the boiling curve, which corresponds to very high surface temperatures. Therefore, any attempt to increase the heat flux beyond $q \cdot \max$ will cause the operation point on the boiling curve to jump suddenly from point C to point E.
- However, surface temperature that corresponds to point E is beyond the melting point of most heater materials, and burnout occurs.
- Therefore, point C on the boiling curve is also called the burnout point, and the heat flux at this point the burnout heat flux (Fig. 10–9).
- Most boiling heat transfer equipment in practice operate slightly below $q \cdot \max$ to avoid any disastrous burnout.
- However, in cryogenic applications involving fluids with very low boiling points such as oxygen and nitrogen, point E usually falls below the melting point of the heater materials, and steady film boiling can be used in those cases without any danger of burnout.



Various boiling regimes during boiling of methanol on a horizontal 1-cm-diameter steam-heated copper tube:

(a) nucleate boiling,

(b) transition boiling

(c) film boiling

(from J. W. Westwater and J. G. Santangelo, University of Illinois at Champaign-Urbana).

Mechanism of nucleate boiling

- In the nucleate boiling regime, the rate of heat transfer strongly depends on the nature of nucleation (the number of active nucleation sites on the surface, the rate of bubble formation at each site, etc.), which is difficult to predict.
- The type and the condition of the heated surface also affect the heat transfer.
- The most widely used correlation for the rate of heat transfer in the nucleate boiling regime was proposed in 1952 by Rohsenow, and expressed as

$$\dot{q}_{\text{nucleate}} = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left[\frac{C_p(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right]^3$$

$\dot{q}_{\text{nucleate}}$ = nucleate boiling heat flux, W/m²

μ_l = viscosity of the liquid, kg/m · s

h_{fg} = enthalpy of vaporization, J/kg

g = gravitational acceleration, m/s²

ρ_l = density of the liquid, kg/m³

ρ_v = density of the vapor, kg/m³

σ = surface tension of liquid–vapor interface, N/m

C_{pl} = specific heat of the liquid, J/kg · °C

T_s = surface temperature of the heater, °C

T_{sat} = saturation temperature of the fluid, °C

C_{sf} = experimental constant that depends on surface–fluid combination

Pr_l = Prandtl number of the liquid

n = experimental constant that depends on the fluid

Mechanism of nucleate boiling

- The rate of heat transfer during nucleate boiling is essentially independent of the geometry and orientation of the heated surface.
- The condition of the heater surface greatly affects heat transfer, and the Rohsenow equation given above is applicable to clean and relatively smooth surfaces.
- The results obtained using the Rohsenow equation can be in error by 100% for the heat transfer rate for a given excess temperature and by 30% for the excess temperature for a given heat transfer rate.
- Therefore, care should be exercised in the interpretation of the results.
- From thermodynamics that the enthalpy of vaporization h_{fg} of a pure substance decreases with increasing pressure (or temperature) and reaches zero at the critical point.
- Noting that h_{fg} appears in the denominator of the Rohsenow equation, we should see a significant rise in the rate of heat transfer at high pressures during nucleate boiling.

Surface tension of liquid–vapor
interface for water

$T, ^\circ\text{C}$	$\sigma, \text{N/m}^*$
0	0.0757
20	0.0727
40	0.0696
60	0.0662
80	0.0627
100	0.0589
120	0.0550
140	0.0509
160	0.0466
180	0.0422
200	0.0377
220	0.0331
240	0.0284
260	0.0237
280	0.0190
300	0.0144
320	0.0099
340	0.0056
360	0.0019
374	0.0

Values of the coefficient C_{sf} and n for various fluid–surface combinations

Fluid-Heating Surface Combination	C_{sf}	n
Water–copper (polished)	0.0130	1.0
Water–copper (scored)	0.0068	1.0
Water–stainless steel (mechanically polished)	0.0130	1.0
Water–stainless steel (ground and polished)	0.0060	1.0
Water–stainless steel (teflon pitted)	0.0058	1.0
Water–stainless steel (chemically etched)	0.0130	1.0
Water–brass	0.0060	1.0
Water–nickel	0.0060	1.0
Water–platinum	0.0130	1.0
<i>n</i> -Pentane–copper (polished)	0.0154	1.7
<i>n</i> -Pentane–chromium	0.0150	1.7
Benzene–chromium	0.1010	1.7
Ethyl alcohol–chromium	0.0027	1.7
Carbon tetrachloride–copper	0.0130	1.7
Isopropanol–copper	0.0025	1.7

Mechanism of film boiling

- Using an analysis similar to Nusselt's theory on filmwise condensation presented in the next section, Bromley developed a theory for the prediction of heat flux for stable film boiling on the outside of a horizontal cylinder.
- The heat flux for film boiling on a horizontal cylinder or sphere of diameter D is given by

$$\dot{q}_{\text{film}} = C_{\text{film}} \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4 C_{pv} (T_s - T_{\text{sat}})]^{1/4}}{\mu_v D (T_s - T_{\text{sat}})} \right] (T_s - T_{\text{sat}})$$

where k_v is the thermal conductivity of the vapor in $\text{W/m} \cdot ^\circ\text{C}$ and

$$C_{\text{film}} = \begin{cases} 0.62 & \text{for horizontal cylinders} \\ 0.67 & \text{for spheres} \end{cases}$$

Forced convection boiling

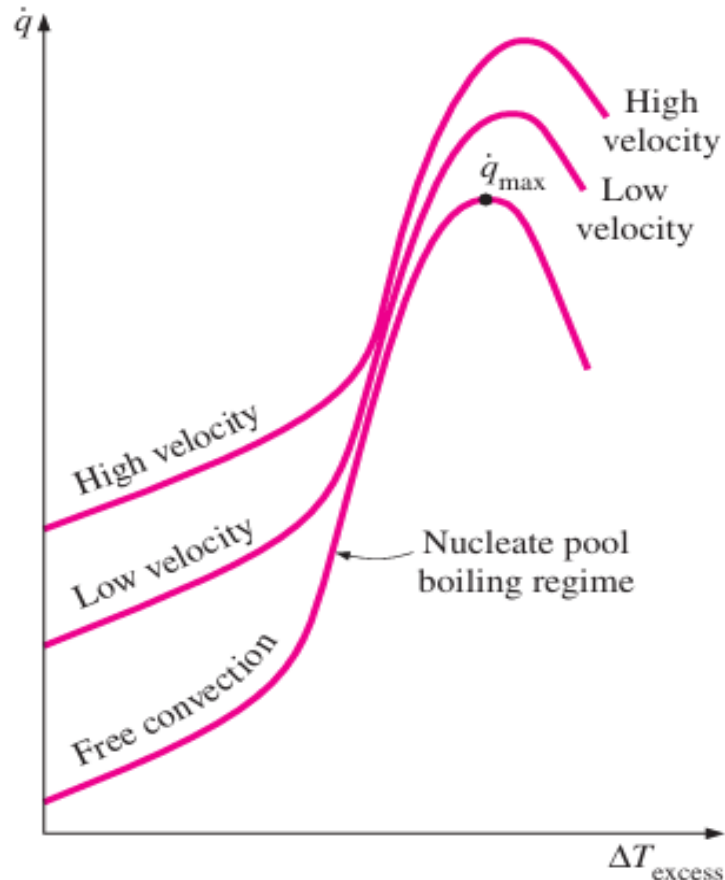
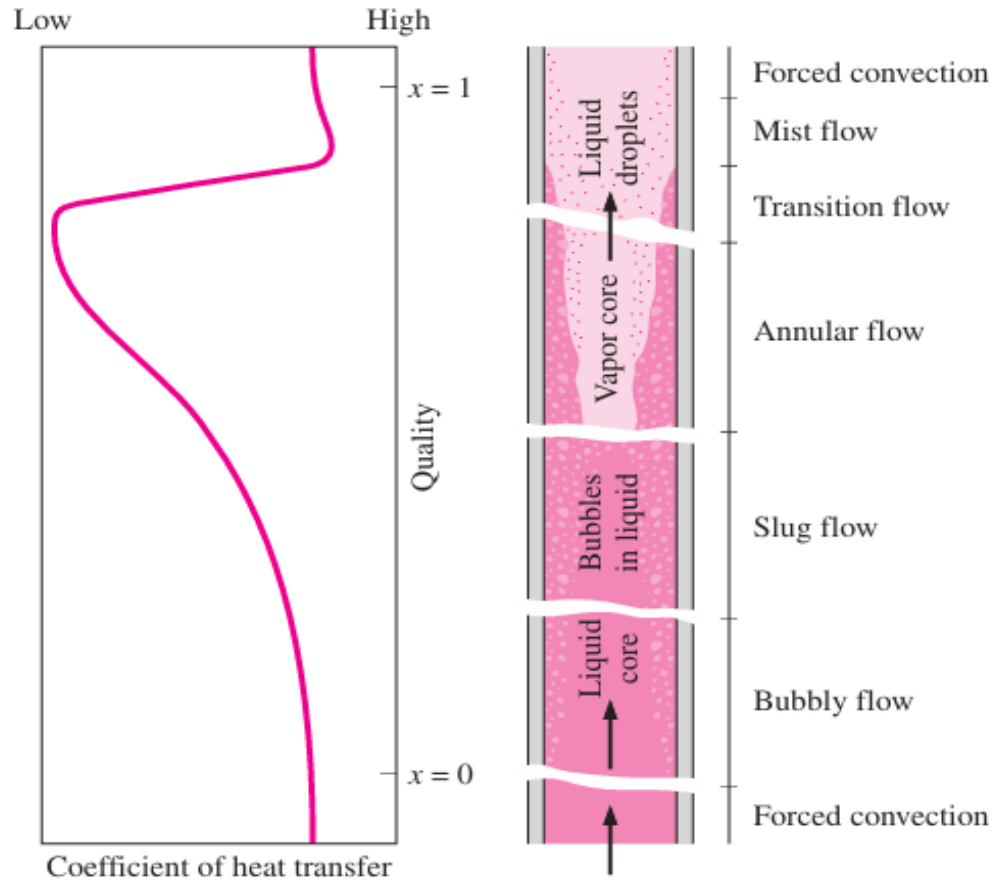


FIGURE 10–18

The effect of forced convection on external flow boiling for different flow velocities.

- In flow boiling, the fluid is forced to move by an external source such as a pump as it undergoes a phase-change process.
- The boiling in this case exhibits the combined effects of convection and pool boiling.
- The flow boiling is also classified as either external and internal flow boiling depending on whether the fluid is forced to flow over a heated surface or inside a heated tube.
- External flow boiling over a plate or cylinder is similar to pool boiling, but the added motion increases both the nucleate boiling heat flux and the critical heat flux considerably.
- Internal flow boiling is much more complicated in nature because there is no free surface for the vapor to escape, and thus both the liquid and the vapor are forced to flow together.
- The two-phase flow in a tube exhibits different flow boiling regimes, depending on the relative amounts of the liquid and the vapor phases. This complicates the analysis even further.

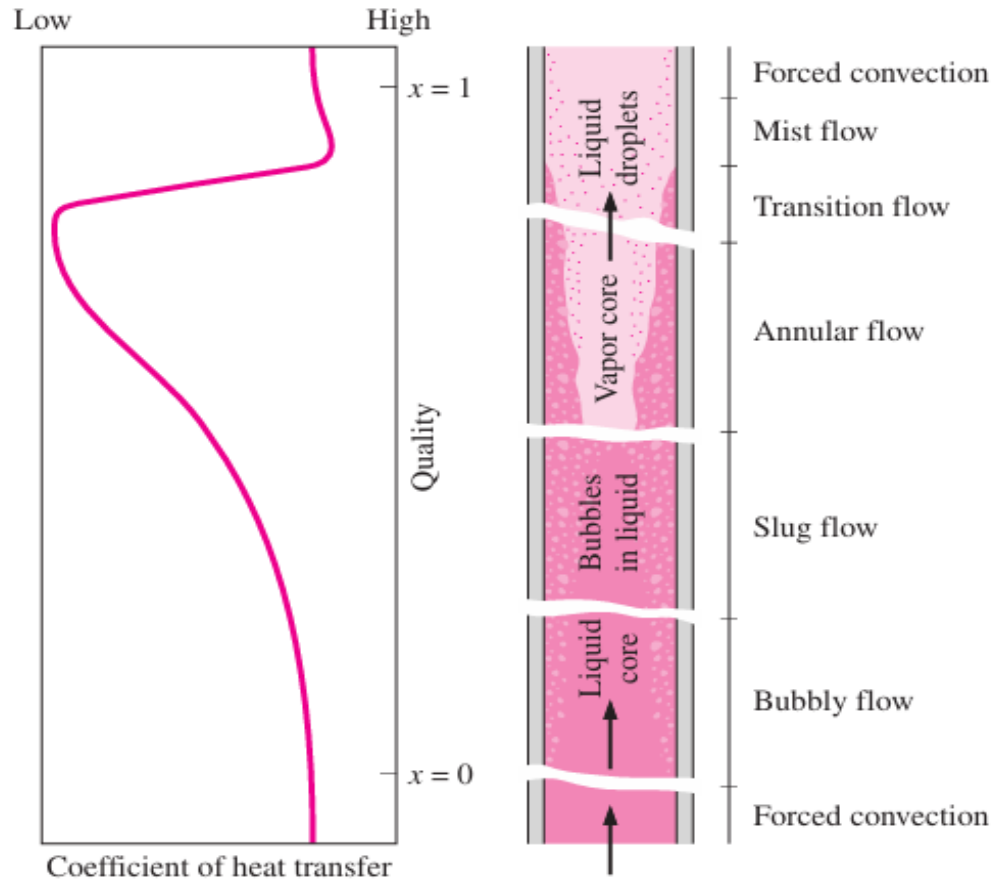
Forced convection boiling



Different flow regimes encountered in flow boiling in a tube under forced convection

- Initially, the liquid is subcooled and heat transfer to the liquid is by forced convection.
- Then bubbles start forming on the inner surfaces of the tube, and the detached bubbles are drafted into the mainstream.
- This gives the fluid flow a bubbly appearance, and thus the name bubbly flow regime.
- As the fluid is heated further, the bubbles grow in size and eventually coalesce into slugs of vapor.
- Up to half of the volume in the tube in this slug flow regime is occupied by vapor.
- After a while the core of the flow consists of vapor only, and the liquid is confined only in the annular space between the vapor core and the tube walls.

Forced convection boiling



Different flow regimes encountered in flow boiling in a tube under forced convection

- This is the annular-flow regime, and very high heat transfer coefficients are realized in this regime.
- As the heating continues, the annular liquid layer gets thinner and thinner, and eventually dry spots start to appear on the inner surfaces of the tube.
- The appearance of dry spots is accompanied by a sharp decrease in the heat transfer coefficient.
- This transition regime continues until the inner surface of the tube is completely dry.
- Any liquid at this moment is in the form of droplets suspended in the vapor core, which resembles a mist, and we have a mist-flow regime until all the liquid droplets are vaporized.
- At the end of the mist-flow regime we have saturated vapor, which becomes superheated with any further heat transfer.

Condensation Heat Transfer

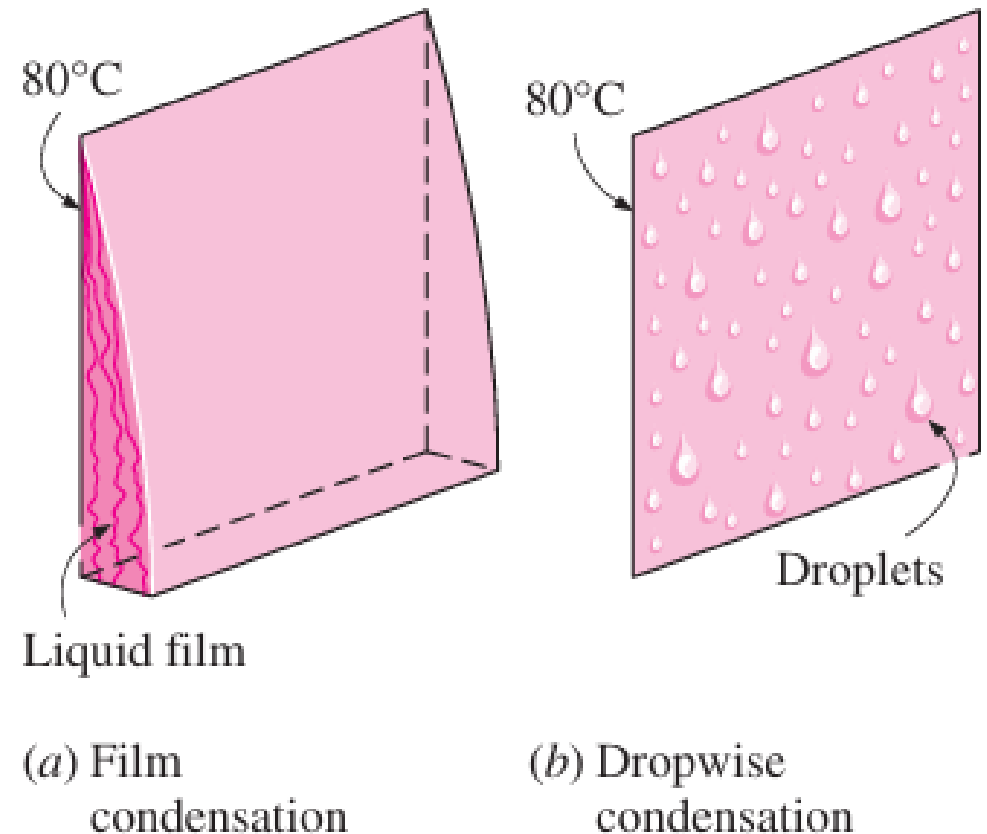
- Condensation occurs when the temperature of a vapor is reduced below its saturation temperature T_{sat} .
- This is usually done by bringing the vapor into contact with a solid surface whose temperature T_s is below the saturation temperature T_{sat} of the vapor.
- But condensation can also occur on the free surface of a liquid or even in a gas when the temperature of the liquid or the gas to which the vapor is exposed is below T_{sat} .
- Two distinct forms of condensation are observed: film condensation and dropwise condensation.
- In film condensation, the condensate wets the surface and forms a liquid film on the surface that slides down under the influence of gravity.
- The thickness of the liquid film increases in the flow direction as more vapor condenses on the film. This is how condensation normally occurs in practice.

Condensation Heat Transfer

- In dropwise condensation, the condensed vapor forms droplets on the surface instead of a continuous film, and the surface is covered by countless droplets of varying diameters. In film condensation, the surface is blanketed by a liquid film of increasing thickness, and this “liquid wall” between solid surface and the vapor serves as a resistance to heat transfer.
- The heat of vaporization h_{fg} released as the vapor condenses must pass through this resistance before it can reach the solid surface and be transferred to the medium on the other side.
- In dropwise condensation, however, the droplets slide down when they reach a certain size, clearing the surface and exposing it to vapor. There is no liquid film in this case to resist heat transfer.
- As a result, heat transfer rates that are more than 10 times larger than those associated with film condensation can be achieved with dropwise condensation.

Condensation Heat Transfer

- Therefore, dropwise condensation is the preferred mode of condensation in heat transfer applications, and people have long tried to achieve sustained dropwise condensation by using various vapor additives and surface coatings.
- These attempts have not been very successful, however, since the dropwise condensation achieved did not last long and converted to film condensation after some time.
- Therefore, it is common practice to be conservative and assume film condensation in the design of heat transfer equipment.



Filmwise Condensation

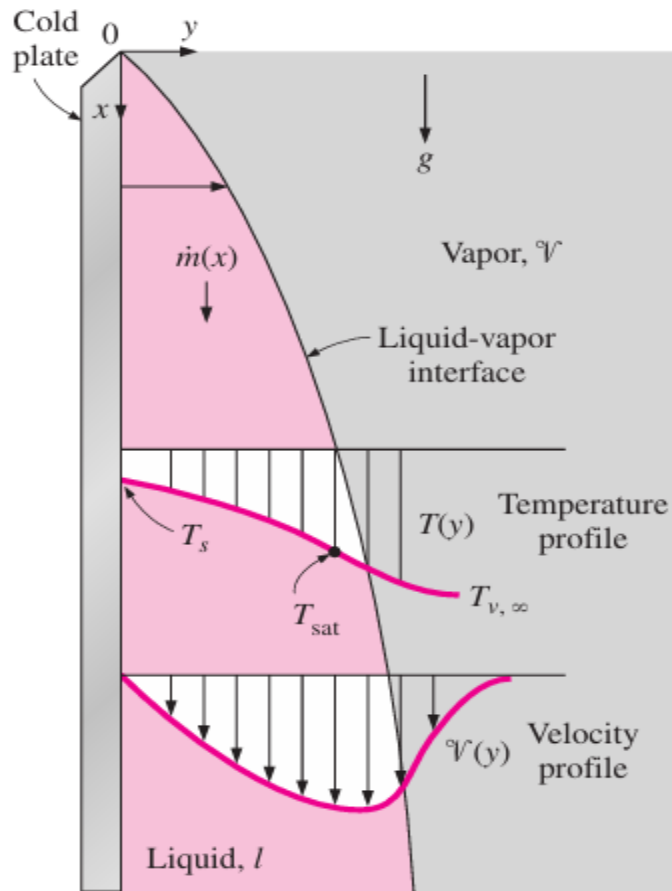


FIGURE 10-21

Film condensation on a vertical plate.

- The liquid film starts forming at the top of the plate and flows down ward under the influence of gravity.
- The thickness of the film increases in the flow direction x because of continued condensation at the liquid–vapor interface.
- Heat in the amount hfg (the latent heat of vaporization) is released during condensation and is transferred through the film to the plate surface at temperature T_s .
- Note that T_s must be below the saturation temperature T_{sat} of the vapor for condensation to occur.
- Note that the velocity of the condensate at the wall is zero because of the “no-slip” condition and reaches a maximum at the liquid–vapor interface.
- The temperature of the condensate is T_{sat} at the interface and decreases gradually to T_s at the wall.
- Heat transfer in condensation also depends on whether the condensate flow is laminar or turbulent.

Film condensation, vertical surfaces, laminar flow

$$Nu = \frac{hL}{k_l} = 1.13 \left(\frac{\rho_l(\rho_l - \rho_v)g\Delta H(T_{sat})L^3}{\mu_l k_l \Delta T} \right)^{\frac{1}{4}} \quad Re = \frac{4m}{\pi D \mu_l} < 1800$$

Equations good for these units:

$$\Delta T [=] K$$

$$h [=] \frac{W}{m^2 K}$$

$$m [=] \frac{kg}{s}$$

$$k_l [=] \frac{W}{m K}$$

$$\rho_v, \rho_l [=] \frac{kg}{m^3}$$

$$\Delta H [=] \frac{J}{kg}$$

$$L [=] m$$

$$\mu_l [=] Pa \cdot s$$

$$g [=] m/s^2$$

$$T_{film} = \frac{T_{wall} + T_{sat}}{2}$$

(All material properties at the film temperature)

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Film condensation, vertical surfaces, turbulent flow

$$Nu = \frac{hL}{k_l} = 0.0077 \left(\frac{\rho_l^2 g L^3}{\mu_l^2} \right)^{\frac{1}{3}} Re^{0.4} \quad Re = \frac{4m}{\pi D \mu_l} > 1800$$

Film condensation, outside horizontal cylinders, laminar flow

$$Nu = \frac{hL}{k_l} = 0.725 \left(\frac{\rho_l(\rho_l - \rho_v)g\Delta H(T_{sat})D^3}{N \mu_l k_l \Delta T} \right)^{\frac{1}{4}} \quad Re = \frac{4m}{\pi D \mu_l} < 1800$$

McCabe W. L., Smith J. C., Harriot P., Unit Operations of Chemical Engineering, 5th edition, McGraw Hill, 1993.

<https://pages.mtu.edu/~fmorriso/cm310/lectures/2015heatlecture09.pdf>

Dropwise Condensation

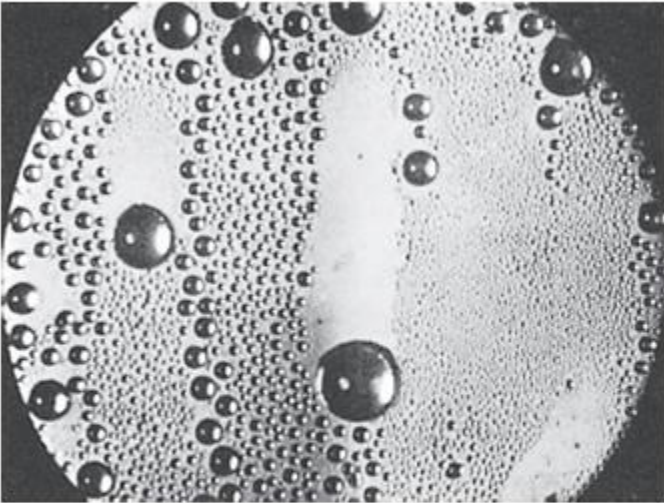


FIGURE 10–35

Dropwise condensation of steam on a vertical surface (from Hampson and Özişik, Ref. 11).

- Dropwise condensation, characterized by countless droplets of varying diameters on the condensing surface instead of a continuous liquid film, is one of the most effective mechanisms of heat transfer, and extremely large heat transfer coefficients can be achieved with this mechanism.
- In dropwise condensation, the small droplets that form at the nucleation sites on the surface grow as a result of continued condensation, coalesce into large droplets, and slide down when they reach a certain size, clearing the surface and exposing it to vapor.
- There is no liquid film in this case to resist heat transfer.
- As a result, with dropwise condensation, heat transfer coefficients can be achieved that are more than 10 times larger than those associated with film condensation.

Dropwise Condensation

- Large heat transfer coefficients enable designers to achieve a specified heat transfer rate with a smaller surface area, and thus a smaller (and less expensive) condenser.
- Therefore, dropwise condensation is the preferred mode of condensation in heat transfer applications. The challenge in dropwise condensation is not to achieve it, but rather, to sustain it for prolonged periods of time.
- Dropwise condensation is achieved by adding a promoting chemical into the vapor, treating the surface with a promoter chemical, or coating the surface with a polymer such as teflon or a noble metal such as gold, silver, rhodium, palladium, or platinum. The promoters
 - used include various waxes and fatty acids such as oleic, stearic, and linoic acids.
 - They lose their effectiveness after a while, however, because of fouling, oxidation, and the removal of the promoter from the surface.
 - It is possible to sustain dropwise condensation for over a year by the combined effects of surface coating and periodic injection of the promoter into the vapor. However, any gain in heat transfer must be weighed against the cost associated with sustaining dropwise condensation.

Filmwise vs. dropwise condensation

Sr. No.	Film condensation	Dropwise condensation
1	The liquid condensates are deposited in the form of a film or layer.	The liquid contents are deposited in the form of droplets of varying sizes.
2	The condensate wets the surface.	It never wets the surface.
3	The film or layer deposited works as insulation by adding extra resistance for heat transfer.	The droplets never wet the surface hence it never acts as insulation.
4	The overall heat transfer coefficient is lower.	The overall heat transfer coefficient is up to 10 times larger than in filmwise condensation.
5	The rate of condensation is lower.	The rate of condensation is higher.
6	The filmwise condensation occurs on less polished and uncontaminated surfaces.	The dropwise condensation requires a highly polished surface or the surface with promoters coating.
7	The size of the equipment for filmwise condensation is larger for the unit rate of condensation.	The equipment is comparatively smaller in size for the unit rate of condensation.